Experiment on Packet Capture tool: Wireshark

Exp no:5 Date:

Aim: Experiments on Packet Capture tool: Wireshark

Packet Sniffer:

→ Sniffs messages being sent/received from by/ your computer

→ Store and display the contents of the various protocol fields in the messages.

→ Passive Program
* Never sends packets itself
* No packets addressed to it.
* Receives a copy of all packets

Packet Sniffer Structure Diagnostic tools:

* Teddump
→ eg: Tcpdump -enx host 10.129.41.2 -w exe3.out

* Wireshark
→ wireshark -r exe3.out

→ Wireshark:

Wireshark, a network analysis tool formely known as Ethereal, captures packets in real time and display them in human-readable format. Wireshark includes filters, color coding, and other features that let you dig deep into network traffic and inspect individual packets.

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Wireshark used for:

* Network administrators, troubleshoot network problem
* Network security engineers: examine security problems
* Define filters - capture and display
* Watch smart statistics

Capturing packets:

→ After downloading and installing Wireshark, launch it and double click the name of a network interface under capture to start capturing packets on that interface.

→ As soon as you click the interface's name, you'll see the packets start to appear in real time. Wireshark captures each packet sent to or from your system.

color coding:

→ You'll probably see packets highlighted in a variety of different colors. Wireshark uses colors to help you identify the types of traffic at a glance. By default, light purple is TCP traffic, light blue is UDP traffic, and black identifies packets with errors, for example, they could have been delivered out of order.

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Filtering Packets:

→ If you're trying to inspect something specific, such as the traffic a program sends when phoning home, it helps to close down all other applications using the network so you can narrow down the traffic.

Inspecting packets:

→ Click a packet to select it and you can dig down to view its details.

→ You can also create filters from here - just right-click one of the details and use the Apply as Filter submenu to create a filter based on it.

Capturing and analysing Packets using Wireshark tool:

→ To filter, capture, view, packets on Wireshark tool. Capture 100 packets from the Ethernet: IEEE 802.3 LAN interface and save it.

Procedure:

* Select Local Area Connection in Wireshark
* Go to capture option
* Select at stop capture automatically after 100 packets
* Then click Start capture
* Save the packets.